Rel-Sortability of Time-Unrolled Causal Graphs: Exact Values, Finite-Sample Saturation, and Pre-Registered Real-Data Tests

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Abstract

Simulated causal-discovery benchmarks can be solved by sorting variables by marginal statistics (var-, R^2- and rel-sortability). Time-unrolled DAGs have recently been proposed as a remedy, with a sufficient condition for their rel-sortability to tend to $1/2$. We give the exact rel-sortability of the time-unrolled DAG for every summary graph, every horizon T and every edge, with an explicit $O(1/T)$ rate. The condition of the earlier result is also necessary, and every non-strongly-connected summary graph, for instance any river network, stays strictly above $1/2$. This excess is caused by the fixed start at $t = 1$: with a stationary start the value is exact for every T and depends only on the source strongly connected components of the summary graph. We show that block-mean aggregation and point subsampling leave this value unchanged but move var- and R^2-sortability to different limits. A finite-sample analysis explains why the population value is hard to see in real series: raw correlation tests are inconsistent under autocorrelation, a latent common driver saturates the relatives count at $n \approx z_c^2/\ell^2$, and proxy conditioning works only with near-perfect reliability and a rank-one driver. All graph-theoretic results and the algebraic core of the finite-sample results are machine-checked in Lean 4 (116 theorems). In three pre-registered real-data studies, a river benchmark fails as the theory predicts, while a physical testbed with designed independent sources gives a partial positive result, including out-of-sample predictions of detection rates.

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1. Introduction

Simulated benchmarks drive much of causal discovery, and several of their regularities can be exploited without any causal reasoning. Marginal variances tend to grow along the causal order (*var-sortability*; Reisach et al., 2021), so a method that sorts by variance does well on data from common generators, and it explains the strong benchmark performance of some continuous-optimisation methods, a family that includes NOTEARS (Zheng et al., 2018). The coefficient of determination R^2 has the same property and is scale invariant (Reisach et al., 2023); internally standardised structural models are not var-sortable by construction and, empirically, mostly not R^2-sortable (Ormaniec et al., 2025); and Herman et al. (2025) relate the R^2 pattern to colliders and confounders and propose a generator that avoids it. Most recently, Reisach et al. (2026) defined *rel-sortability*, which orders a node by the number of its *relatives* (nodes joined to it by an open path, marginally dependent on it under faithfulness). Relatives are a property of the graph alone, so the criterion cannot be removed by reweighting. For random DAGs it is high; for *time-unrolled* DAGs built from a summary graph with self-loops, Reisach et al. (2026, Thm. 4) show that it tends to $1/2$ if every weakly connected component of the summary graph is strongly connected, and they propose such graphs as benchmarks.

Three questions remain open, and they matter for real time series, whose summary graphs are often feed-forward (rivers, supply chains, signalling cascades) and whose observations are aggregated or subsampled: (Q1) what is the rel-sortability of a time-unrolled DAG when the summary graph is not strongly connected; (Q2) does it depend on how the series is started and observed; and (Q3) can a relatives count be estimated from one finite, autocorrelated record, so that a method could actually exploit it?

Contributions. (1) The exact rel-sortability of every edge of the time-unrolled DAG G_T, for every summary graph and every T (Theorem 1), its limit $\frac{1}{2} + |E_\times|/(2|E_S|)$ with $E_\times$ the edges between strongly connected components, and an explicit $O(1/T)$ rate (Theorem 2). The condition of Reisach et al. (2026, Thm. 4) is also necessary (Corollary 3). The excess above $1/2$ for, e.g., chains and out-trees is a boundary effect of the deterministic start at $t = 1$ (Proposition 4).

(2) A stationary version (latent burn-in of $p-1$ slices or an infinite past) whose value is exact for every T and is fixed by the *source* strongly connected components (Theorem 5); river networks exceed $1/2$ iff they contain a confluence (Corollary 6). **(3)** Under temporal aggregation the relatives are unchanged, whereas var- and R^2-sortability converge to those of a long-run structural model under block means and stay at the snapshot values under subsampling (Section 4). **(4)** Finite-sample theory (F1–F3, Section 5): a recovery bound with an effective sample size and the inconsistency of the nominal-n test under autocorrelation; saturation by a latent driver at $n \approx z_c^2/\ell^2$; and a proxy-conditioning analysis showing that the proxy needs reliability close to 1 and the driver must be one-dimensional. **(5)** Machine-checked proofs in Lean 4 with Mathlib of all graph results and of the deterministic core of F1–F3 (116 theorems, standard axioms only; Section 6). Every theorem below names its Lean counterpart. **(6)** Three pre-registered real-data studies (Section 7). On CausalRivers (Stein et al., 2025) the relatives sort carries no orientation information, and F2/F3 predict the observed dependence rates. On the Causal Chambers (Gamella et al., 2025) two of three pre-registered criteria pass, a light-tunnel replication passes all three, and F1/F2 predict held-out detection rates within 0.02. We report the failed pre-registered tests in the main text.

The graph results of contributions (1)–(2) are short consequences of a walk-length lemma and of Reisach et al. (2026, Lemma 1). Their value is that they settle the question completely, are machine-checked, and change what a benchmark designer should conclude: much of the rel-sortability of time-unrolled benchmarks is an artefact of how the simulation is started. The finite-sample results use textbook tools; their value is that they predicted, quantitatively and in one case out of sample, the outcome of real-data tests.

Related work. Beyond the sortability papers above, Lohse & Wahl (2025) measure var- and R^2-sortability of real and simulated time series and note that cyclic pairs push the signal towards $1/2$; they give no theorem on relatives or aggregation. Identifiability from subsampled (Gong et al., 2015) and aggregated (Gong et al., 2017) series is a separate question from sortability. Time-series discovery methods such as PCMCI (Runge et al., 2019) test conditional independences, and our finite-sample results concern only the marginal tests that define relatives.

2. Setting

For a directed graph, $\mathrm{anc}(y)$ and $\mathrm{desc}(y)$ contain y; the relatives of y are $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{anc}(y)) = \{z : \mathrm{anc}(y) \cap \mathrm{anc}(z) \neq \emptyset\}$ and $\rho(y) = |\mathrm{rel}(y)|$. For an edge (x, y), $s_\rho(x, y) = 1, \frac{1}{2}, 0$ if $\rho(x) <, =, > \rho(y)$, and $s_\rho(G)$ is the mean over edges (Reisach et al., 2026). A value $1/2$ means the criterion carries no orientation information.

A *summary graph* $G_\mathrm{S} = (V, E_\mathrm{S})$ has $p \geq 2$ nodes and a self-loop at every node. The *time-unrolled DAG* G_T has nodes $v(t)$, $t \in \{1, \ldots, T\}$, and an edge $(i(t), j(t{+}1))$ for each $(i, j) \in E_\mathrm{S}$ and $t < T$, so every summary edge has $T-1$ copies. $d(u, v)$ is the shortest-path length in G_S (self-loops ignored; $d(v, v) = 0$; ∞ if unreachable), and anc_S, rel_S are taken in G_S. SCC means strongly connected component. $E_\times \subseteq E_\mathrm{S}$ is the set of edges between different SCCs and $E_\circ = E_\mathrm{S} \setminus E_\times$. A *source* SCC has no incoming edge from another SCC, and $\mathrm{Src}(v)$ is the set of source SCCs that reach v.

3. Exact Rel-Sortability of Time-Unrolled DAGs

Theorem 1 (every edge, every T)**.** *For $(i, j) \in E_\mathrm{S}$ and $1 \leq t < T$, the edge $(i(t), j(t{+}1))$ of G_T has $s_\rho = 1$ if some $u \in V$ satisfies $d(u, j) \leq t \leq d(u, i)$, and $s_\rho = 1/2$ otherwise. The value does not depend on T.*

Lean: `edge_score_dist`, `edge_score`, `strictCond_iff_dist`.

Sketch. Every edge of G_T advances time by one, so $u(s)$ is an ancestor of $v(t)$ iff $d(u, v) \leq t - s$ (pad a shortest path with self-loops). Hence the roots of $v(t)$ are $\{u(1) : d(u, v) \leq t - 1\}$. Since $i(t)$ is a parent of $j(t{+}1)$, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the inclusion is strict iff $j(t{+}1)$ has a root that $i(t)$ lacks (Reisach et al., 2026, Thm. 1, Lemma 1). Full proof in Appendix A. (The Lean proof characterises rel in G_T directly and does not use their Lemma 1.)

Theorem 2 (exact limit and rate)**.** *For every $T \geq 2$, with $L(G_\mathrm{S}) = \frac{1}{2} + \frac{|E_\times|}{2|E_\mathrm{S}|}$,*

$$0 \leq s_\rho(G_T) - L(G_\mathrm{S}) \leq \frac{\min(p{-}1, T{-}1)\,|E_\circ|}{2(T-1)|E_\mathrm{S}|} \leq \frac{p-1}{2(T-1)},$$

so $\lim_{T\to\infty} s_\rho(G_T) = L(G_\mathrm{S})$.

Lean: `sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_GT`.

Sketch. For a crossing edge, j does not reach i; take $u = j$ in Theorem 1: all $T-1$ copies score 1. For an edge inside an SCC, i and j are reached by the same nodes, which forces $t \leq d(u, i) \leq p - 1$, so at most $p-1$ copies score 1.

Corollary 3 (necessary and sufficient condition)**.** *$\lim_T s_\rho(G_T) = 1/2$ iff every weakly connected component of G_S is strongly connected. Otherwise $s_\rho(G_T) \geq \frac{1}{2} + \frac{1}{2|E_\mathrm{S}|}$ for every T. The "if" half is Reisach et al. (2026, Thm. 4). If G_S is acyclic apart from self-loops, with m other edges,*

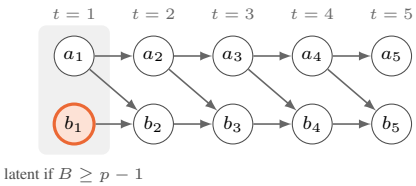


Figure 1. Summary chain $a \to b$ unrolled to $T = 5$. Every a_t has the relatives $\{a_s\} \cup \{b_s : s \geq 2\}$ ($2T - 1$ nodes); every b_t with $t \geq 2$ also has b_1 ($2T$ nodes). So each edge $a_t \to b_{t+1}$ is strictly sorted (finite limit $L = 2/3$), but only because of the root b_1 at the fixed start. With a latent burn-in slice, both counts are $2T$ and $s_\rho^W = 1/2$.

$L = \frac{1}{2} + \frac{m}{2(m+p)}$ *whatever its shape (chain, in-tree, out-tree, collider).*

Lean: `tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`.

Proposition 4 (the excess is a boundary effect)**.** *For $t \geq p$, $\rho(v(t)) = T\,|\mathrm{rel}_S(v)| - \sum_{w \in \mathrm{rel}_S(v)} d(\mathrm{anc}_S(v), w)$.*

Lean: `rho_unrolled_large_t`, `rho_unrolled_eq`.

So for a crossing edge with $\mathrm{rel}_S(i) = \mathrm{rel}_S(j)$ (root to child in an out-tree, a chain), the strict order rests on $O(1)$ extra relatives out of $\Theta(T)$, all located in the first p slices. Figure 1 shows the smallest case. A sample-based relatives count on a long record cannot see such a difference, which motivates the stationary version.

Theorem 5 (stationary start)**.** *Observe the window of the last $T \geq 2$ slices of G_{B+T} with a latent burn-in $B \geq p - 1$ (or of the infinite-past graph on $V \times \mathbb{Z}$), with relatives computed in the full graph. Then every window node has $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$, and for every T*

$$s_\rho^W = \tfrac{1}{2} + \frac{|E_\uparrow|}{2|E_S|},\ E_\uparrow = \{(i,j) \in E_S : \mathrm{Src}(i) \neq \mathrm{Src}(j)\}.$$

Lean: `rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `relSet_eq_iff_src`.

Sketch. After $p-1$ slices every finite distance is absorbed, so $w(s) \in \mathrm{rel}(v(t))$ iff $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$. For an edge (i,j), $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ with equality iff $\mathrm{Src}(i) = \mathrm{Src}(j)$.

Corollary 6 (stationary condition; graph classes)**.** $s_\rho^W = 1/2$ *iff every weakly connected component of G_S has exactly one source SCC. Strongly connected graphs, out-forests, chains and forks give $1/2$. In an in-forest (a river network), $s_\rho^W = \frac{1}{2} + \#\{\text{edges into confluences}\}/(2|E_S|)$, which exceeds $1/2$ iff there is a confluence.*

Lean: `sBurn_eq_half_iff`, `sZ_eq_half_iff`, `stat_half_of_outForest`, `statValue_inForest`, `statValue_inForest_gt_iff`, `collider_stat`, `fork_stat`.

Table 1. Rel-sortability of 3-node summary graphs (with self-loops, $|E_S| = 5$). Finite: $\lim_T s_\rho(G_T)$ (Theorem 2); stationary: s_ρ^W (Theorem 5). Lean: `chain3_limit`, `chain3_stat`, `fork_limit`, `fork_stat`, `collider_limit`, `collider_stat`; 3-cycle by `reisach_theorem4`.

summary graph	finite G_T (limit)	stationary
chain $a \to b \to c$	0.7	0.5
fork $a \leftarrow b \to c$	0.7	0.5
collider $a \to c \leftarrow b$	0.7	0.7
3-cycle (6 edges)	0.5	0.5

Table 1 shows the two versions side by side. In the finite generator every summary DAG with the same m/p has the same value, so a benchmark built this way rewards relatives-sorting on chains and forks, where the stationary process gives no signal. For a discovery method applied to long stationary records, the stationary value is the relevant one, and it is what our real-data predictions use. We verified all statements of this section by exhaustive enumeration of all summary graphs with $p \leq 4$ and 10^4 random graphs with $p \leq 7$ (46,724 (graph, T) cases, 0 failures; Appendix B) before the Lean proofs were written.

4. Aggregation versus Subsampling

Let X be a stable linear SVAR with lags $\tau \geq 1$, independent innovations with diagonal covariance Σ, and summary graph G_S; $\Gamma(h)$ is its autocovariance and $A(1) = \sum_\tau A_\tau$. Consider block means $\bar{Y}_s = \frac{1}{D}\sum_{u<D} X(sD+u)$ and subsamples $Z_s = X(sD)$.

Proposition 7 (relatives survive any per-node filter)**.** *If $\mathrm{anc}_S(i) \cap \mathrm{anc}_S(j) = \emptyset$ then X_i and X_j are independent at all lags, and so are any per-node, time-invariant (possibly nonlinear) transforms of them, including $\bar{Y}$ and Z for every D. If the ancestor sets intersect, $\Gamma(0)_{ij}$, $\mathrm{Cov}(\bar{Y})_{ij}$ and the long-run covariance are nonzero for Lebesgue-almost every parameter.*

Lean: not machine-checked (needs the SVAR moving-average representation and real-analytic genericity).

So the population relatives of contemporaneous values are rel_S for every observation scheme, and the population relatives sort equals the stationary value of Theorem 5 (on non-self-loop edges), not the finite-G_T value. Var- and R^2-sortability behave differently.

Theorem 8 (block means vs subsampling)**.** *(a) $\|D\,\mathrm{Cov}(\bar{Y}) - S_{LR}\| \leq \frac{1}{D}\sum_h |h|\|\Gamma(h)\|$, with $S_{LR} = (I - A(1))^{-1}\Sigma(I - A(1))^{-\top}$. (b) If G_S is acyclic apart from self-loops, S_{LR} is the covariance of a static SEM on G_S with weights $A(1)_{ji}/(1-\lambda_j)$ and noise variances $\sigma_j^2/(1-\lambda_j)^2$, $\lambda_j = A(1)_{jj} < 1$. (c) $\mathrm{Cov}(Z_s) = \Gamma(0)$ for every D. (d) Hence, for large D, block-mean R^2-sortability is 0 on out-stars and 1 on in-stars with ≥ 2 arms, for all weights; under subsampling*

every contemporaneous criterion keeps its $\Gamma(0)$ value.

Lean: (a) counting identity `fejer_identity`; (b) `longrun_factor`, `selfloop_sum_lt_one`; the probabilistic steps, the rate and (c)–(d) are not machine-checked.

Block means and subsampling therefore converge to different contemporaneous objects (a long-run SEM versus the stationary snapshot), and block means can reverse the R^2 order a snapshot has: in our exact computations on out-stars, subsampled data are R^2-sorted correctly on average (0.84) while block means drift to 0 (Appendix C). Rel-sortability is the only one of the three criteria that depends on the graph alone and is invariant to the observation scheme, which is why we study it further.

5. Finite-Sample Theory

We now ask whether the population relatives can be estimated. X is a zero-mean stationary Gaussian process with summable autocovariances $\gamma_{ab}(h)$, standardised so that $\gamma_{aa}(0) = 1$, observed at n times. Two nodes are declared relatives if the Fisher-z test of the sample correlation $\hat{r}_{ab}$ rejects at level α with Bonferroni over $m = d(d-1)/2$ pairs, i.e. $|\hat{r}_{ab}| > \tanh(z_c/\sqrt{n'-3})$, $z_c = \Phi^{-1}(1-\alpha/2m)$; $\hat{\rho}(v) = 1 + \#\text{declared partners}$. The *nominal* test uses $n' = n$. Let $B_{ab} = \sum_h |\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)|$ (for an independent pair, the Bartlett sum $\sum_h \rho_a(h)\rho_b(h)$).

F1: recovery and effective sample size.

Theorem 9 (recovery bound)**.** *For $x, y > 0$ let $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$, $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n < 1$ and $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$, with autocovariance constants $D_\gamma, \kappa_1, \kappa_2$ defined in Appendix D. If thresholds separate ($\delta_{ab} \le \tau_{ab}$ on non-relatives and $|\rho_{ab}| - \delta_{ab} > \tau_{ab}$ on relatives), the declared set equals the true one, and hence $\hat{s} = s_\rho^{\mathrm{pop}}$, with probability at least $1 - d(d-1)e^{-x} - 4de^{-y}$. For a common threshold $\tau < \rho_{\min}$ the failure probability decays as $\exp(-n\min(\tau, \rho_{\min} - \tau)^2/(2(\sqrt{B} + \rho_{\max}\sqrt{D_\gamma})^2))$, so n/B plays the role of an effective sample size.*

Lean: deterministic core `thm15_det`, `decide_eq_target`, `corr_perturb`, `sOf_cntDecl_congr`; Chernoff and trace algebra `chernoff_choice`, `phi_le`, `lemma12_trace`, `lemma12_le_B`. The Gaussian tail bound itself is not machine-checked.

For i.i.d. snapshots $B = 1 + \rho^2$; under strong persistence B can be in the tens, and so can the loss in effective sample size. The proof combines a Gaussian-chaos tail bound (the Gaussian case of Hsu et al., 2012, Prop. 1; see also Laurent & Massart, 2000), an exact finite-n variance identity for $\hat{r}$, and a deterministic perturbation lemma.

Proposition 10 (the nominal test is inconsistent; asymptotic)**.** *Assuming the standard CLT for quadratic forms of stationary Gaussian processes, a non-relative pair is declared by the nominal test with probability $\to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$, which does not vanish. With $n_{\mathrm{eff}} = n(1-\rho^2)^2/W_{ab}$ (W the delta-method Bartlett variance) the false-positive rate tends to α/m.*

Lean: not machine-checked (relies on a CLT not available in Mathlib).

F2: saturation by a latent driver. Let $X_v = Y_v + b_v C$ with C a stationary unit-variance process independent of the VAR Y, and $\ell_v = b_v^2/\mathrm{Var}X_v$ the driver's share of variance.

Proposition 11 (saturation)**.** *For summary non-relatives, $\rho_{uv} = \mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v} =: \kappa_{uv}$. If all $b_v \neq 0$, every pair is a population relative and the population rel-sortability is exactly $1/2$; $P(\hat{s}_n \neq \frac{1}{2}) \le (d^2+3d)e^{-cn}$ with $c \approx \kappa_{\min}^2/(18(\sqrt{B} + \sqrt{D_\gamma})^2)$. Asymptotically, a pair is detected with probability $> 1/2$ once $n > n_{1/2} = 3 + (z_c/\operatorname{atanh}\kappa)^2 \approx z_c^2/\ell^2$ (nominal test; for the effective test, the same bound on n_{eff}).*

Lean: `prop21_corr_std`, `sOf_saturated`, `abs_sOf_sub_half_le`, location of $n_{1/2}$: `ncrit_eq_iff`; the probability statements are not machine-checked.

$n_{1/2}$ does not depend on autocorrelation, which only sets the width of the transition. A driver with share $\ell = 0.1$ saturates at $n \approx 10^3$, and one with $\ell = 0.64$ at $n \approx 25$. Since $\tau_n \to 0$ for any consistent test, a relatives count on a long record can recover the summary relatives only inside a window $\max_{\text{non-rel}} |\kappa| < \tau_n < \min_{\text{rel}} |\rho|$, which closes at n of order $n_{1/2}$. Autocorrelation (Proposition 10) and a latent driver are two separate failure modes; both push $\hat{s}$ towards $1/2$.

F3: conditioning on a proxy. Let $Z = \sqrt{r}\,C + \sqrt{1-r}\,e$ with reliability $r = \mathrm{corr}(Z, C)^2$, and let π_{uv} be the partial correlation given Z.

Proposition 12 (conditioning)**.** *With ρ^Y the driver-free correlation,*

$$\pi_{uv} = \frac{\rho^Y_{uv}\sqrt{(1-\ell_u)(1-\ell_v)} + (1-r)\,\mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v}}{\sqrt{(1-r\ell_u)(1-r\ell_v)}}.$$

At $r = 1$ the conditional relatives are the summary relatives and Theorem 5 is restored. For non-relatives with $\ell_u, \ell_v \in (0,1)$, $|\pi_{uv}|$ is strictly decreasing in r and vanishes only at $r = 1$. Spurious conditional relatives therefore appear for every $r < 1$ once n is large; fewer than half of the non-relative pairs are declared only if $r > r_{\mathrm{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares). If the driver reaches nodes through different filters (e.g. travel times k_v), a scalar proxy leaves a nonzero residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$.

Lean: `prop31_std`, `cor32`, `absPi_strictAntiOn`, `absPi_eq_zero_iff`, `rcrit_iff`, `thm35_bracket`,

`pc_lipschitz`, `prop36_ar1_ne_zero`, `prop36_rank_one`, `prop38_ratio`.

Formalisation found four missing nondegeneracy hypotheses in our own earlier statements (e.g. $b_u b_v \neq 0$ for strict monotonicity); they are included above and listed in Appendix F. For $\ell = 0.64$ and a typical threshold $\tau' = 0.082$, $r_{\text{crit}} = 0.95$.

The non-asymptotic bounds are valid but loose (factors 3–13 in n in simulation, Appendix E); the asymptotic statements matched every simulation to within 0.03 and are what we use for prediction below.

6. Machine-Checked Proofs

All results of Section 3 and the deterministic and algebraic parts of Sections 4 and 5 are formalised in Lean 4 (de Moura & Ullrich, 2021) with Mathlib (The mathlib Community, 2020) (3,343 lines). The axiom report lists 116 theorems, each depending only on `propext`, `Classical.choice` and `Quot.sound`; the sources contain no `sorry`, `admit` or `axiom`. Relatives, s_ρ, G_T, burn-in windows and the infinite-past graph are defined from scratch following Reisach et al. (2026); the one modelling convention is that "a directed path exists" is read as reflexive-transitive reachability. Not formalised: Gaussian tail bounds, Isserlis' theorem, the CLT, the SVAR moving-average representation and genericity (Proposition 7). These are the parts a referee must check by hand; Appendix F maps every statement to its Lean name and status.

7. Pre-Registered Real-Data Evidence

Protocol. We ran three studies. Each prediction, threshold, estimator, seed and kill rule was committed to version control before any dependence statistic of the data was computed (timeline in Appendix H). The predicted value for an observed node set S is the stationary value (Theorem 5) computed in the full known graph with unobserved nodes latent, averaged over the induced non-self-loop edges ($\tilde{s}$). The estimator is the relatives count of Section 5 at $\alpha = 0.01$ with Bonferroni; orientation accuracy is the mean edge score of $\hat{\rho}$. CIs are cluster bootstraps (2,000 resamples).

CausalRivers: a negative result the theory explains. River networks are in-forests, so Corollary 6 predicts $\tilde{s} > 1/2$ for subgraphs with confluences (mean 0.82). The estimated relatives sort carried no orientation information (P2, P3 fail; Table 2) because 98% of non-relative station pairs tested dependent. Read through F2 (post hoc for this study): non-relative correlations $\kappa \approx 0.64$–0.74 give $n_{1/2} \approx 20$–27, while the records have $n = 377$–$3{,}652$ steps; the study sat two orders of magnitude inside the saturated regime (Figure 2), where Proposition 11 predicts $\hat{s} = 1/2$. Regional

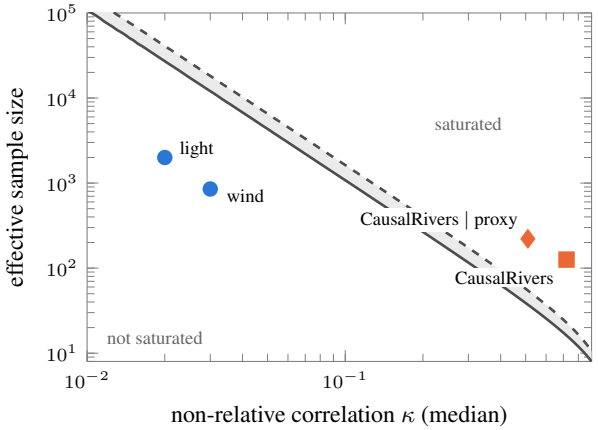


Figure 2. Saturation regime of Proposition 11. Curves: onset $n_{1/2}$ for $z_c = 3.29$ ($d = 5$, solid) and 4.04 ($d = 20$, dashed); above them a non-relative pair is detected with probability $> 1/2$. Points: medians from the real-data studies (CausalRivers unconditioned and given the regional proxy, κ replaced by $|\pi|$; Causal Chambers wind and light tunnel). Values are in Section 7 and Appendix G.

precipitation acts as a latent driver; adding it to the summary graph leaves one source per component, and Corollary 6 then predicts exactly $1/2$. The second study pre-registered the F3 remedy (condition on a leave-out regional mean). It failed (P4: 0.74), and F3 accounts for it: the triad estimate of reliability is $\hat{r} \approx 0.62$, below the $r_{\text{crit}} \approx 0.87$ required at the observed n_{eff} (only 26% of pairs exceed it), and the plug-in prediction of the post-conditioning dependent fraction is 0.743 (0.736) against an observed 0.741 (0.735) at 12 h (24 h). This check is in-sample, because $\hat{r}$ and $\hat{\ell}$ come from the tested correlations. Post hoc, conditioning on 10 principal components still leaves 49% of non-relative pairs dependent, consistent with a driver of rank > 1 (Proposition 12).

Causal Chambers: designed independent sources. The third study chose data in which Corollary 6's non-trivial case holds by design: in the wind-tunnel random-walk runs (Gamella et al., 2025) five actuators are independent experimenter-generated random walks, so they are separate source SCCs; 25 of 26 edges are predicted to be strictly sorted. Saturation is absent (P7, 0.039) and the relatives count orients confluence edges above chance (P8, 0.656). The graded agreement P9 missed its threshold ($0.267 < 0.3$; CI contains 0.3), so by our own pre-registered definition the study is *not* a positive real-data demonstration. The reason is F1, not a driver: 16 of the 26 ground-truth edges are essentially undetectable at lag 0 at $n \approx 10^3$ (smallest relative correlation ≈ 0.002), which compresses the estimated counts. The light-tunnel replication with i.i.d. inputs passes all three criteria (Table 2). F-theory predictions computed from the first half of each unit (plug-in correlations and autocorrelations, Fisher-z asymptotics of Propositions 10 and 11) and evaluated on the second half are accurate to within 0.02 (Table 3). In particular F2 predicts no saturation:

Table 2. Pre-registered tests. Pass/fail is by the pre-registered point rule. Brackets: 95% cluster-bootstrap CI. "non-rel. dep.": fraction of graph non-relative pairs declared dependent. "Acc.": orientation accuracy on edges predicted to be strictly sorted.

study (units)	test (threshold)	result (12 h / 24 h)	verdict
CausalRivers, marginal (120 subgraphs)	P1: confluence subgraphs have $\tilde{s} > 1/2$	112 of 114	fails as worded[a]
	P2: Acc. > 0.5, CI excl. 0.5	$0.501\,[0.492, 0.509]$ / $0.498\,[0.489, 0.507]$	**fail**
	P3: Spearman$(\tilde{s}, \text{acc.}) \geq 0.3$	-0.018 / 0.018	**fail**
	(non-rel. dep., nominal test)	0.984 / 0.978	kill rule triggered
CausalRivers, conditioned on regional proxy (120 subgraphs)	P4: non-rel. dep. < 0.5 given proxy	0.741 / 0.735	**fail**
	P5: Acc. > 0.5, CI excl. 0.5	$0.539\,[0.513, 0.565]$ / $0.540\,[0.514, 0.567]$	pass, not interp.[b]
	P6: Spearman ≥ 0.3	-0.113 / -0.121	**fail**
Causal Chambers, wind tunnel (15 runs, primary)	P7: non-rel. dep. < 0.5	$0.039\,[0.031, 0.047]$	pass
	P8: Acc. > 0.5, CI excl. 0.5	$0.656\,[0.637, 0.675]$	pass
	P9: Spearman ≥ 0.3 (300 subgraphs)	$0.267\,[0.163, 0.368]$	**fail**
light tunnel (10 segments, replication)	R7 / R8 / R9 (same thresholds)	0.000 / 1.000 / $0.779\,[0.696, 0.864]$	pass (replication)[c]

[a]The two exceptions are split-and-rejoin channels with a single source, for which the theorem itself predicts $1/2$. [b]Kill rule (P4 failed at both resolutions): P5 is reported, not interpreted. [c]Pre-registered as not counting towards a positive demonstration (i.i.d. inputs, weak time-series aspect).

Table 3. Out-of-sample F-theory check on the Causal Chambers: predictions from the first half of each unit, observed rates on the second half. Accuracy is predicted by Monte Carlo of independent pair decisions.

	wind tunnel		light tunnel	
	pred.	obs.	pred.	obs.
non-relatives declared (F2)	0.031	0.028	0.006	0.000
relatives detected (F1)	0.360	0.343	0.679	0.681
orientation accuracy	0.630	0.620	0.999	1.000

$\hat{\kappa} \approx 0.03$ (wind) and 0.02 (light) give $n_{1/2} \approx 15{,}600$ and 33,800, far above the available $n_{\text{eff}} \approx 430$–$1{,}000$.

A measured driver (ambient pressure, shared by four barometers with no graph edge) is detected exactly where F2 predicts ($\kappa = 0.19$–0.32, $n_{1/2} = 145$–$426 < n_{\text{eff}}$), and conditioning on it removes it (dependent fraction $0.039 \to 0.001$). Two caveats: the wind tunnel graph is bipartite, so the prediction coincides with "sensors have more relatives than actuators"; and all units of a chamber share one device.

E1: discriminating test against naive predictors. **[PLACEHOLDER: E1 results from the experiment currently running. Intended content: whether the theorem-predicted $\tilde{s}$ predicts estimated accuracy better than naive graph predictors (to address the bipartite caveat above). State pre-registration commit, design, thresholds, verdicts. No numbers until the run is complete.]**

E2: lagged relatives. **[PLACEHOLDER: E2 results from the experiment currently running. Intended content: whether a relatives test that uses lagged cross-correlations recovers the weak lag-0 edges that limit P9 (F1 caveat). State pre-registration, design, verdicts. No numbers until the run is complete.]**

8. Discussion and Limitations

For benchmark design. The finite time-unrolled generator is rel-sortable above $1/2$ for every summary graph that is not a union of strongly connected components (Corollary 3), and part of the excess comes only from the start at $t = 1$. Benchmarks should either use strongly connected summary graphs or a burn-in of at least $p - 1$ slices, after which the value is the graph property of Theorem 5, for every T. Reporting $|E_\times|/|E_\text{S}|$ or $|E_\uparrow|/|E_\text{S}|$ alongside a benchmark costs nothing.

For real data. Population rel-sortability above $1/2$ does not imply that a relatives count can be estimated. Two conditions are needed, and F1–F3 quantify both: the absence of a latent common driver with share ℓ such that $n \gtrsim z_c^2/\ell^2$ (or a near-perfect, rank-one proxy for it), and relatives that are detectable at lag 0 at the available n_{eff}. CausalRivers violates the first, the wind tunnel partly the second. A cheap diagnostic follows: before relying on relatives-based orderings (or before reading a benchmark result as evidence that a method uses them), estimate the typical correlation $\hat{\kappa}$ of pairs believed to be unrelated and compare n_{eff} with $3 + (z_c/\operatorname{atanh}\hat{\kappa})^2$; if n_{eff} is far above it, the relatives count will be saturated whatever the graph. A practical consequence for method developers is that a time-series discovery method cannot gain much from rel-sortability on real records with a shared driver, even when the summary graph would make it informative in population.

Open questions. Summary graphs without self-loops, contemporaneous effects at the native time step, and nonlinear mechanisms (where Proposition 7 still gives the zero pattern but faithfulness can fail, as the Malus-law pairs of the light tunnel show) are not covered. A test that uses lagged cross-correlations could recover the weak lag-0 relatives

that limited P9 (experiment E2).

Limitations. The graph results assume self-loops at every node (without them a single edge gives $1/2$ instead of $2/3$; Appendix A). Section 4 covers linear SVARs only. The finite-sample bounds are loose, and the asymptotic statements rest on a CLT that we cite rather than prove. Proposition 7 and the probabilistic parts of F1–F3 are proved on paper only. The real-data evidence comes from two benchmark families, and the positive part from one physical device per chamber.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Artefact Statement

Lean code, verification scripts, analysis scripts, preregistration documents and run logs are provided as supplementary material (anonymised). Raw data are public (CausalRivers; Causal Chambers, CC BY 4.0) and are not redistributed.

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Appendix overview

Appendix A gives full proofs of Section 3, Appendix B the exhaustive enumeration check, Appendix C the proofs of Section 4, Appendix D the proofs of Section 5, Appendix E the simulation checks of F1–F3, Appendix F the map from statements to Lean names, Appendix G details of the real-data studies, Appendix H the pre-registration timeline and deviations, and Appendix I the placeholders for experiments E1 and E2.

Status labels. *Machine-checked*: a Lean theorem with no `sorry` and only the standard axioms. *Proved*: a complete paper proof given here. *Asymptotic*: the argument uses the standard central limit theorem for quadratic forms of stationary Gaussian processes with summable covariances, which we cite and do not prove.

A. Proofs for Section 3

Throughout, $G_{\mathrm{S}} = (V, E_{\mathrm{S}})$ has a self-loop at every node, $p = |V| \geq 2$, and $d(u, v)$ is as in Section 2. We use two facts of Reisach et al. (2026), proved there for DAGs: (Thm. 1) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subseteq \mathrm{rel}(y)$; (Lemma 1, eq. (10)) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subsetneq \mathrm{rel}(y)$ iff $\mathrm{roots}(y) \setminus \mathrm{roots}(x) \neq \emptyset$, where $\mathrm{roots}(y) = \{x \in \mathrm{anc}(y) : \mathrm{anc}(x) = \{x\}\}$. Both are also machine-checked independently (`relSet_subset_of_edge`, and the direct characterisation `isRel_unrolled_iff`).

Lemma 13 (ancestors in G_T)**.** *$u(s) \in \mathrm{anc}(v(t))$ iff $s \leq t$ and $d(u, v) \leq t - s$.*

Lean: `unrolled_reach_iff`, `unrolled_reach_iff_dist`.

Proof. Every edge of G_T increases time by one, so a directed path from $u(s)$ to $v(t)$ has $t - s$ edges, and its projection to V is a walk of length $t - s$ from u to v in G_{S} (self-loops allowed as steps). Such a walk exists iff $d(u, v) \leq t - s$: a walk contains a path, so its length is at least $d(u, v)$; conversely, pad a shortest path with $t - s - d(u, v)$ self-loop steps at u. Every such walk lifts to a directed path in G_T. □

Lemma 14 (roots)**.** $\mathrm{roots}(v(t)) = \{u(1) : d(u, v) \leq t - 1\}$, *and* $\mathrm{desc}(u(1)) = \{w(s) : d(u, w) \leq s - 1\}$.

Lean: `roots_unrolled_iff`, `isRoot_unrolled_iff`, `desc_root_unrolled_iff`.

Proof. A node $u(s)$ with $s \geq 2$ has the parent $u(s - 1)$ through the self-loop, so it is not a root; $u(1)$ has no parent. Apply Lemma 13 with $s = 1$, in both directions. □

Proof of Theorem 1. $i(t)$ is a parent of $j(t{+}1)$, so $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$ and $s_\rho(e) \in \{1/2, 1\}$, with value 1 iff the inclusion is strict, iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$. By Lemma 14 this set is $\{u(1) : d(u, j) \leq t,\ d(u, i) > t - 1\}$, and since d is integer-valued or ∞, $d(u, i) > t - 1$ iff $d(u, i) \geq t$. □

Proof of Proposition 4 (and the general count). Since every ancestor of y descends from a root of y that is itself an ancestor, $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{roots}(y))$. By Lemma 14, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u, v) \leq t - 1$ and $d(u, w) \leq s - 1$, i.e. iff $s \geq 1 + m_t(v, w)$ with $m_t(v, w) = \min\{d(u, w) : d(u, v) \leq t - 1\}$. Counting $s \in \{1, \ldots, T\}$ gives $\rho(v(t)) = \sum_w \max(T - m_t(v, w), 0)$ (`rho_unrolled_eq`). For $t \geq p$ every finite $d(u, v)$ is at most $p - 1 \leq t - 1$, so the admissible u are $\mathrm{anc}_S(v)$ and $m_t(v, w) = d(\mathrm{anc}_S(v), w)$, finite iff $w \in \mathrm{rel}_S(v)$ and then at most $p - 1 < t \leq T$. (The Lean statement `rho_unrolled_large_t` needs only $t \geq p$; an earlier hypothesis $T \geq p$ is redundant.) □

Proof of Theorem 2. Crossing edge $(i, j) \in E_\times$: j does not reach i, so $d(j, i) = \infty$, and $u = j$ gives $d(j, j) = 0 \leq t \leq \infty$ for every t: all $T - 1$ copies score 1. Edge in $E_\circ$: i and j reach each other (or $i = j$), so for every u, $d(u, i) < \infty$ iff $d(u, j) < \infty$. If the condition of Theorem 1 held with $d(u, i) = \infty$ then $d(u, j) = \infty > t$, a contradiction; hence it forces $t \leq d(u, i) \leq p - 1$, and at most $N_T(i, j) \leq \min(p - 1, T - 1)$ copies score 1. Summing over the $(T - 1)|E_{\mathrm{S}}|$ edges,

$$s_\rho(G_T) = \tfrac{1}{2} + \frac{1}{2(T-1)|E_{\mathrm{S}}|}\Big[(T-1)|E_\times| + \sum_{(i,j)\in E_\circ} N_T(i, j)\Big],$$

which gives both bounds; let $T \to \infty$. □

Proof of Corollary 3. $E_\times = \emptyset$ iff no edge joins two SCCs. A weakly connected component made of two or more SCCs is connected by undirected edges, one of which joins two different SCCs; conversely a strongly connected component has no

crossing edge. The uniform bound is Theorem 2 with $|E_\times| \geq 1$. For the DAG case, every SCC is a singleton, $E_\times$ consists of the m non-loop edges, and $|E_{\mathrm{S}}| = m + p$. □

Proof of Theorem 5. In G_{B+T}, by Lemma 14 shifted by B, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \leq t-1$ and $d(u,w) \leq s-1$. For window nodes $t-1, s-1 \geq B \geq p-1$, and every finite distance is at most $p-1$, so the condition is $u \in \mathrm{anc}_S(v) \cap \mathrm{anc}_S(w)$, independent of s,t. Hence $\mathrm{rel}_W(v(t)) = \{w(s) \in W : w \in \mathrm{rel}_S(v)\}$, of size $T|\mathrm{rel}_S(v)|$. On $V \times \mathbb{Z}$, Lemma 13 holds verbatim and the common ancestor can be taken arbitrarily early, giving the same set. Each copy of (i,j) therefore compares $|\mathrm{rel}_S(i)|$ with $|\mathrm{rel}_S(j)|$; as $i \in \mathrm{anc}_S(j)$, $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$, so the score is 1 iff $\mathrm{rel}_S(i) \neq \mathrm{rel}_S(j)$, and $1/2$ for self-loops. Source SCCs: $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i) \neq \emptyset$ iff $\mathrm{anc}_S(j) \not\subseteq \mathrm{rel}_S(i)$ (if $z \in \mathrm{rel}_S(i)$ then $\mathrm{desc}_S(z) \subseteq \mathrm{rel}_S(i)$). Every node of $\mathrm{anc}_S(j)$ is reached from some $\sigma \in \mathrm{Src}(j)$, and $\mathrm{desc}_S(\sigma) \subseteq \mathrm{rel}_S(i)$ whenever $\sigma \subseteq \mathrm{rel}_S(i)$. A source SCC meets $\mathrm{rel}_S(i)$ iff it reaches i (its only ancestors are its own nodes), and then it is contained in $\mathrm{rel}_S(i)$. So the condition is $\mathrm{Src}(j) \not\subseteq \mathrm{Src}(i)$, which, as $\mathrm{Src}(i) \subseteq \mathrm{Src}(j)$, is $\mathrm{Src}(i) \neq \mathrm{Src}(j)$. □

Proof of Corollary 6. If each weak component C has a unique source SCC σ_C, then $\mathrm{Src}(v) = \{\sigma_C\}$ on C and $E_\uparrow = \emptyset$. Conversely, if $E_\uparrow = \emptyset$, Src is constant along edges and hence on each weak component; a node v of a source SCC σ has $\mathrm{Src}(v) = \{\sigma\}$, so there is only one. Out-forests (at most one parent besides itself; acyclicity is not needed, `stat_half_of_outForest`) and strongly connected graphs (`stat_half_of_strong`) have one source SCC per component. In an in-forest (acyclic apart from self-loops, out-degree at most one), if $u \in \mathrm{anc}(v)$ then $\mathrm{desc}(u)$ is the unique downstream path through v, so $\mathrm{rel}(v) = \mathrm{anc}(v) \cup \mathrm{desc}(v)$ (`inForest_isRel_iff`); the source SCCs are the headwaters, and $\mathrm{Src}(j) = \mathrm{Src}(i) \cup \bigcup_k \mathrm{Src}(k)$ over the other parents k of j, with disjoint headwater sets on different branches, so $\mathrm{Src}(j) \neq \mathrm{Src}(i)$ iff j is a confluence (`inForest_up_iff`). Out-degree at most one alone is not enough: for $a \rightleftarrows b$ with $c \to a$, a has two parents but the only source SCC is $\{c\}$. □

Remark 15 (self-loops). Without self-loops the lemmas fail: for the single edge $0 \to 1$ without loops, $s_\rho(G_T) = 1/2$ for every $T \geq 2$ (`edgeNoLoops_sRho`); with loops the limit is $2/3$ (`edgeLoops_limit`).

B. Exhaustive Enumeration Check

Before formalisation, every statement of Section 3 was checked by brute force (bitset relatives with the definitions of Reisach et al., 2026) on all $4 + 64 + 4096$ summary graphs with $p = 2, 3, 4$ (all $T \leq 9$) and on 10,000 random graphs ($p = 2$–7; Erdős–Rényi, DAG, in-forest, out-forest, chain with back edge and block-SCC families; three values of $T \leq 14$ each, fixed seed). In total 46,724 (graph, T) cases and 1,895,609 edges were checked with zero failures of Theorem 1, Proposition 4, the bounds of Theorem 2, Theorem 5 (with $B = p-1$ and $B = 2p+1$) and the source-SCC characterisation. The conjecture we had pre-registered ("an edge is strictly sorted in the limit iff $|\mathrm{rel}_S(j)| > |\mathrm{rel}_S(i)|$") disagreed with the exact G_T limit in 9,603 random graphs and never with the stationary value; the conjecture is Theorem 5, not the limit of G_T.

C. Proofs for Section 4

Assumptions: $X(t) = \sum_{\tau=1}^{L} A_\tau X(t-\tau) + \varepsilon(t)$, $t \in \mathbb{Z}$, i.i.d. innovations with diagonal positive-definite covariance Σ, stable (all roots of $\det(I - \sum_\tau A_\tau z^\tau)$ outside the unit disc), stationary solution; the summary graph has $i \to j$ iff $(A_\tau)_{ji} \neq 0$ for some τ.

Proof of Proposition 7. By the MA(∞) representation $X(t) = \sum_k \Psi_k \varepsilon(t-k)$, and because Ψ_k is supported on the reachability relation, $X_i(t)$ is a measurable function of $\{\varepsilon_k(s) : k \in \mathrm{anc}_S(i)\}$. Disjoint ancestor sets mean independent innovation sets, so the processes X_i and X_j are independent, and so are any per-node transforms. For the generic part, each covariance entry is real-analytic in (A, Σ) on the connected open set of stable parameters with the given support, hence either identically zero or nonzero almost everywhere. With all coefficients positive, including the self-loop at a common ancestor k, all Ψ entries are nonnegative and $\Psi_{ik}(m), \Psi_{jk}(m) > 0$ for all large m (self-loops pad the walks), so $\Gamma(0)_{ij} > 0$; the same holds for $\Gamma(h)$, $h \geq 0$, and therefore for the Fejér sums and S_{LR}. "Generic" does not mean "large": with little autocorrelation, $\Gamma(0)_{ij}$ along a lagged edge can be tiny (down to 10^{-14} in our exact computations; exactly 0 without autocorrelation). □

Proof of Theorem 8. (a) $D^2 \mathrm{Cov}(\bar{Y}) = \sum_{u,w<D} \Gamma(u-w) = D\Gamma(0) + \sum_{h=1}^{D-1} (D-h)(\Gamma(h) + \Gamma(-h))$ (the counting identity is `fejer_identity`). The stable VAR has $\|\Gamma(h)\| \leq C r^{|h|}$, $r < 1$, so $\sum_h |h| \|\Gamma(h)\| < \infty$ and $S_{LR} =$

$\sum_h \Gamma(h) = \Psi(1)\Sigma\Psi(1)^\top$ with $\Psi(1) = (I - A(1))^{-1}$. Then $S_{LR} - D\,\mathrm{Cov}(\bar{Y}) = \sum_{|h|\geq D}\Gamma(h) + \sum_{|h|<D}\frac{|h|}{D}\Gamma(h)$, and $1 \leq |h|/D$ in the first sum. (b) Write $I - A(1) = (I-\Lambda) - N$ with Λ the diagonal and N the off-diagonal part; then $I - A(1) = (I-\Lambda)(I-\tilde{B})$ with $\tilde{B} = (I-\Lambda)^{-1}N$ strictly triangular in a topological order (`longrun_factor`, `longrun_inv`). Under acyclicity the AR polynomial factorises into univariate $\phi_j(z) = 1 - \sum_\tau (A_\tau)_{jj} z^\tau$; stability, $\phi_j(0) = 1$ and the intermediate value theorem give $\phi_j(1) = 1 - \lambda_j > 0$ (`selfloop_sum_lt_one`). (c) Stationarity. (d) For a linear SEM $X = BX + e$ with noise variances ω, $R_i^2 = 1 - 1/(\Sigma_{ii}(\Sigma^{-1})_{ii})$ and $(\Sigma^{-1})_{ii} = 1/\omega_i + \sum_{k\in\mathrm{ch}(i)} B_{ki}^2/\omega_k$. In an out-star with root 0 and leaves k, $1 - R_0^2 = 1/(1+\sum_k r_k)$ and $1 - R_k^2 = 1/(1+r_k)$ with $r_k = \omega_0 B_{k0}^2/\omega_k > 0$, so with ≥ 2 leaves the root has strictly larger R^2 than every leaf; the in-star is symmetric. Strict inequalities in the limit SEM of (b), plus (a) and continuity of $S \mapsto R^2$ at positive-definite S, give the claim for all large D. Under subsampling every contemporaneous statistic is a function of $\Gamma(0)$. □

The R^2 statement for stars strengthens the three-node analysis in Herman et al. (2025, App. D), whose qualitative message (that R^2 reflects connectivity, with an asymmetry between colliders and confounders) we credit. Numerical observations (exact population moments, not proved): mean block-mean R^2-sortability over 150 in-trees moves from 0.853 ($D = 1$) to 0.834 ($D = 4$–8) and 0.880 ($D = 256$), so it is not monotone in D; on out-stars subsampled data are R^2-sorted at 0.842 on average while block means give 0.546 ($D = 4$), 0.101 ($D = 16$) and 0 in the limit.

D. Proofs for Section 5

Notation. $\hat{C}_{ab} = \frac{1}{n}\sum_t (X_a(t) - \bar{X}_a)(X_b(t) - \bar{X}_b)$, $\hat{r}_{ab} = \hat{C}_{ab}/\sqrt{\hat{C}_{aa}\hat{C}_{bb}}$. Constants: B_{ab} as in the main text; $D_\gamma = \max_a 2\sum_h \gamma_{aa}(h)^2$; $\kappa_1 = \max_a \sum_h |\gamma_{aa}(h)|$; $\kappa_2 = \max_{a\neq b}(\sum_h |\gamma_{aa}(h)| + \sum_h |\gamma_{ab}(h)|)$. For i.i.d. sampling, $B_{ab} = 1 + \rho_{ab}^2$, $D_\gamma = 2$, $\kappa_1 = 1$.

Lemma 16 (Gaussian chaos tail)**.** *Let $g \sim N(0, I_N)$, A symmetric, $S = g^\top Ag - \mathrm{tr}A$. For $x > 0$, $P(S \geq 2\|A\|_F\sqrt{x} + 2\|A\|_{op}x) \leq e^{-x}$, and the same for $-S$.*

Lean: algebraic steps `phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`; the probability statement is not machine-checked.

Proof. Diagonalise; S has the law of $\sum_i \lambda_i(g_i^2 - 1)$. For $2s\lambda < 1$, $\log Ee^{s\lambda(g^2-1)} = \varphi(s\lambda)$ with $\varphi(u) = -\frac{1}{2}\log(1-2u) - u = \sum_{k\geq 2}(2u)^k/(2k) \leq u^2/(1-2|u|)$ for $|u| < 1/2$. With $c = 2\|A\|_{op}$, $v = 2\|A\|_F^2$ and $0 \leq s < 1/c$, $\log Ee^{sS} \leq vs^2/(2(1-cs))$. Markov's inequality with $t = \sqrt{2vx} + cx$, $q = \sqrt{2x/v}$ and $s = q/(1+cq)$ gives exponent exactly x. This is the Gaussian case of Hsu et al. (2012, Prop. 1), who derive it from Laurent & Massart (2000); the proof here is self-contained. □

Lemma 17 (exact variance)**.** *With $\tilde{\gamma}_{ab} = \frac{1}{n}\sum_t X_a(t)X_b(t) = g^\top Ag$ for $A = C^{1/2}MC^{1/2}$, $M = \frac{1}{2n}\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$,*

$$2n\|A\|_F^2 = \sum_{|h|<n}\left(1 - \tfrac{|h|}{n}\right)\left[\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)\right] \leq B_{ab},$$

$\|A\|_{op} \leq \kappa_2/(2n)$; on the diagonal $2n\|A\|_F^2 \leq D_\gamma$ and $\|A\|_{op} \leq \kappa_1/n$; and $\bar{X}_a \sim N(0, \bar{\sigma}_a^2)$ with $\bar{\sigma}_a^2 \leq \kappa_1/n$.

Lean: trace and Toeplitz algebra `quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `lemma12_le_B`, `lemma12_diag_le_D`; the step $\|C^{1/2}MC^{1/2}\|_F^2 = \mathrm{tr}(MCMC)$ and the operator-norm bounds are not machine-checked.

Proof. $\|A\|_F^2 = \mathrm{tr}(MCMC)$ and, with Toeplitz blocks $(C_{ab})_{st} = \gamma_{ab}(s-t)$, $4n^2\mathrm{tr}(MCMC) = 2\sum_{s,t}[\gamma_{aa}(s-t)\gamma_{bb}(s-t) + \gamma_{ab}(s-t)\gamma_{ab}(t-s)]$ (using that γ_{aa}, γ_{bb} are even); count pairs with $s - t = h$. $\|A\|_{op} \leq \|C\|_{op}\|M\|_{op}$ and $\|C\|_{op}$ is at most the largest absolute row sum. □

Lemma 18 (deviation event)**.** *Let $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$ and $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n$. With probability at least $1 - d(d-1)e^{-x} - 4de^{-y}$, $|\hat{C}_{ab} - \rho_{ab}| \leq \varepsilon_{ab}$ for all $a \neq b$ and $|\hat{C}_{aa} - 1| \leq \varepsilon_d$ for all a.*

Lean: deterministic steps `centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`.

Proof. $\hat{C}_{ab} = \tilde{\gamma}_{ab} - \bar{X}_a\bar{X}_b$. Apply Lemmas 16 and 17 to each off-diagonal and diagonal $\tilde{\gamma}$, bound $|\bar{X}_a| \leq \sqrt{2\kappa_1 y/n}$ by the Gaussian Chernoff bound, and take a union bound. □

Lemma 19 (correlation perturbation). *If $a', b' \in [1-\varepsilon_d, 1+\varepsilon_d]$, $\varepsilon_d < 1$ and $|c'-\rho| \le \varepsilon$, then $|c'/\sqrt{a'b'} - \rho| \le (\varepsilon + |\rho|\varepsilon_d)/(1-\varepsilon_d)$.*

Lean: `corr_perturb` *(the hypothesis $|\rho| \le 1$ of our earlier statement is not needed).*

Proof. $|c'/\sqrt{a'b'} - \rho| \le |c'-\rho|/\sqrt{a'b'} + |\rho||1/\sqrt{a'b'} - 1|$, with $1/\sqrt{a'b'} \le 1/(1-\varepsilon_d)$ and $|1/\sqrt{a'b'} - 1| \le \varepsilon_d/(1-\varepsilon_d)$. □

Proof of Theorem 9. Set $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. On the event of Lemma 18, Lemma 19 gives $|\hat{r}_{ab} - \rho_{ab}| \le \delta_{ab}$, so non-relatives satisfy $|\hat{r}| \le \tau$ and relatives $|\hat{r}| > \tau$ (`decide_eq_target`, `thm15_det`). Equal declared sets give equal counts and hence $\hat{s} = s_\rho^{\mathrm{pop}}$ (`sOf_cntDecl_congr`). For a VAR observed as a long stationary series, s_ρ^{pop} is the stationary value of Theorem 5 on non-self-loop edges (Proposition 7). Explicit form: with $y = x + \log(4/(d-1))_+$ (`explicit_y`) and a common threshold $\tau < \rho_{\min}$, $P(\hat{R} \ne R) \le 2d(d-1)\exp\big(-n\min(\tau, \rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2)(1+o(1))\big)$. □

Argument for Proposition 10 (asymptotic). By Isserlis' theorem, $n\,\mathrm{Cov}(\tilde{\gamma}_{ab}, \tilde{\gamma}_{cd}) \to V(ab,cd) = \sum_h [\gamma_{ac}(h)\gamma_{bd}(h) + \gamma_{ad}(h)\gamma_{bc}(h)]$ (as in Lemma 17); centring is $O(1/n)$. Assuming the CLT for quadratic forms, the delta method with gradient $(-\rho/2, -\rho/2, 1)$ gives $\sqrt{n}(\hat{r} - \rho) \Rightarrow N(0, W)$, $W = V(ab,ab) - \rho[V(ab,aa) + V(ab,bb)] + \frac{\rho^2}{4}[V(aa,aa) + 2V(aa,bb) + V(bb,bb)]$, which equals B_{ab} for an independent pair. Then $\sqrt{n}\,\operatorname{atanh}\hat{r} \Rightarrow N(0, B_{ab})$ under the null and $P(|\operatorname{atanh}\hat{r}| > z_c/\sqrt{n-3}) \to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$. □

Proof of Proposition 11. $\mathrm{Cov}(X_u, X_v) = \gamma_{uv}^Y(0) + b_u b_v$ and $\gamma_{uv}^Y = 0$ for non-relatives (Proposition 7); divide by the standard deviations (`prop21_corr_std`). For relatives the sum is nonzero except on a Lebesgue-null set. All counts equal d, so every edge scores $1/2$ (`sOf_saturated`). The rate: apply Theorem 9 with target "all pairs" and $x = y = cn$, c small enough that $\delta \le \kappa_{\min}/3$ and $\tau_n \le \kappa_{\min}/3$ eventually; $|\hat{s} - 1/2| \le 1/2$ always (`abs_sOf_sub_half_le`). The onset: by the asymptotic normality of $\operatorname{atanh}\hat{r}$, the declaration probability crosses $1/2$ where $\tanh(z_c/\sqrt{n-3}) = |\kappa|$, i.e. at $n_{1/2}$ (`ncrit_eq_iff`). □

Proof of Proposition 12. $\mathrm{Cov}(X_u, Z) = b_u\sqrt{r}$ and $\mathrm{Var}Z = 1$; the partial covariance is $\gamma_{uv}^Y + (1-r)b_u b_v$ and the partial variance $\mathrm{Var}Y_u + (1-r)b_u^2$; divide and standardise (`prop31`, `prop31_std`; positive $\mathrm{Var}Y_u$ is required). At $r = 1$, $\pi = \rho^Y$ (`cor32`). For non-relatives $|\pi_{uv}| = \sqrt{\ell_u \ell_v}\sqrt{f_u f_v}$ with $f_w(r) = (1-r)/(1-r\ell_w)$ and $f_w'(r) = (\ell_w - 1)/(1-r\ell_w)^2 < 0$ (`absPi_strictAntiOn`, `absPi_eq_zero_iff`; both need $b_u b_v \ne 0$). Bounds $(1-r)\sqrt{\ell_u\ell_v} \le |\pi| \le (1-r)\sqrt{\ell_u\ell_v}/(1-\ell_{\max})$ (`absPi_lower`, `absPi_upper`). r_{crit} solves $(1-r)\ell/(1-r\ell) = \tau'$ (`rcrit_iff`, `rcrit_example`). The finite-sample bracket uses the Lipschitz bound $|g(a',b',c') - g(a,b,c)| \le L(q)\max(|a'-a|, |b'-b|, |c'-c|)$ for $g(a,b,c) = (a-bc)/\sqrt{(1-b^2)(1-c^2)}$, $|b|, |c| \le q < 1$, $L(q) = 1/(1-q^2) + 4q/(1-q^2)^2$ (`pc_lipschitz`, `thm35_bracket`), combined with Lemma 18 for (X, Z). Multi-filter driver: for $D_v(t) = b_v C(t - k_v)$ with C an AR(1) with coefficient ϕ and $Z = C(t)$,

$$\pi_{uv} = \frac{b_u b_v(\phi^{|k_u - k_v|} - \phi^{k_u + k_v})}{\sqrt{(s_u^2 - b_u^2\phi^{2k_u})(s_v^2 - b_v^2\phi^{2k_v})}},$$

nonzero for $k_u, k_v > 0$, $\phi \in (0,1)$, $b_u b_v \ne 0$ (`prop36_ar1`, `prop36_ar1_ne_zero`); in general $\pi_{uv} \propto \mathrm{Cov}(D_u - P_Z D_u, D_v - P_Z D_v)$ vanishes for all non-relative pairs iff the residual driver components are uncorrelated, which holds for a rank-one driver with $Z = F$ (`prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`). A proxy that contains summary relatives of u, v adds the terms $-c_Z(b_u q_v + b_v q_u) - q_u q_v$ to the numerator (`prop37_pcov`, `prop37_bound`). Triad identity: $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ and, for $r > 0$, $\ell_u = \rho_{uZ}^2/r$ (`prop38_ratio`, `prop38_loading`). □

E. Simulation Checks of F1–F3

All with fixed seeds; full tables are in the supplementary material. **F1** (5 processes: in-forests with $d = 5$ and $d = 10$, self-loop persistence $\lambda \in \{0.5, 0.9, 0.97\}$, and a near-white-noise case; $n = 200$–$20{,}000$; 1,000 replications per cell; nominal, oracle-effective and fixed thresholds). The bound was never violated in 105 cells. It is loose: with a fixed threshold $\rho_{\min}/2$, it reaches 0.9 at about 3–13 times the n at which the simulated probability does, and much later with Bonferroni thresholds, where the Chernoff step loses the Mills-ratio factor. The empirical $n\,\mathrm{Var}(\hat{r})/W$ lies in $[0.93, 1.11]$. With the nominal test, $P(\hat{R} = R)$ plateaus at 0.94, 0.22, 0.03 and 0.27, as Proposition 10 predicts (predicted vs simulated false-positive rates, e.g. 0.33 vs 0.32–0.33 at $\lambda = 0.9$). **F2** ($\ell \in \{0.02, 0.1, 0.3, 0.64\}$): the asymptotic detection curve matches the simulated fraction of declared non-relatives within 0.02 for $n \ge 200$; predicted $n_{1/2} = 1{,}079$ for $\ell = 0.1$,

simulated 0.479 at $n = 1{,}000$. **F3**: predicted $n_{\text{crit}} = 472$ ($\ell = 0.64$, $r = 0.9$) gives a simulated declared fraction 0.495 at $n = 471$; over 38 cells the asymptotic prediction is within 0.03. With travel-time lags $(0, 1, 2, 4, 8)$, $\phi = 0.9$, $r = 1$, the declared fraction still rises to 0.63 at $n = 10{,}000$, as the multi-filter part of Proposition 12 predicts.

F. Lean Formalisation: Map and Findings

Lean 4.35.0 and Mathlib v4.35.0. The build log of a clean rebuild, the `#print axioms` output for each of the 116 main theorems (all `[propext, Classical.choice, Quot.sound]`) and a grep showing no `sorry`/`admit`/`axiom` are in the supplementary material.

statement	Lean names	status
Lemmas 13 and 14	`unrolled_reach_iff`, `unrolled_reach_iff_dist`, `roots_unrolled_iff`, `desc_root_unrolled_iff`, `isRel_unrolled_iff`	checked
Theorem 1	`edge_score`, `edge_score_dist`, `strictCond_iff_dist`	checked
Proposition 4	`rho_unrolled_eq`, `mT_le_iff`, `rho_unrolled_large_t`	checked
Theorem 2	`sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_unrolled`, `tendsto_sRho_GT`	checked
Corollary 3	`tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`	checked
Theorem 5	`rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `sOf_window`, `relSet_eq_iff_src`, `mem_upEdges_iff`	checked
Corollary 6	`sBurn_eq_half_iff`, `sBurn_eq_half_iff_common_anc`, `sZ_eq_half_iff`, `upEdges_eq_empty_tfae`, `stat_half_of_strong`, `stat_half_of_outForest`, `inForest_isRel_iff`, `inForest_up_iff`, `statValue_inForest`, `statValue_inForest_gt_iff`	checked
Table 1	`collider_limit`, `fork_limit`, `chain3_limit`, `collider_stat`, `fork_stat`, `chain3_stat`	checked
self-loop remark	`edgeLoops_limit`, `edgeNoLoops_sRho`	checked
Proposition 7	–	paper only
Theorem 8(a)	`fejer_identity` (counting identity)	partial
Theorem 8(b)	`longrun_factor`, `longrun_inv`, `selfloop_sum_lt_one`	partial
Theorem 8(c,d)	–	paper only
Lemma 16	`phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`	algebra only
Lemma 17	`quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `avg_toeplitz_le_tsum`, `lemma12_le_B`, `lemma12_diag`, `lemma12_diag_le_D`	algebra only
Lemma 18	`centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`	deterministic part
Lemma 19	`corr_perturb`	checked
Theorem 9	`decide_eq_target`, `decide_eq_support`, `thm15_det`, `explicit_y`, `sOf_cntDecl_congr`	deterministic part
Proposition 10	–	asymptotic
Proposition 11	`prop21_corr`, `prop21_corr_std`, `sOf_saturated`, `sOf_const`, `abs_sOf_sub_half_le`, `thm22_const`, `ncrit_iff`, `ncrit_eq_iff`	deterministic part
Proposition 12	`pcorr_eq_pcov`, `pcorr_eq_corr_resid`, `prop31`, `prop31_std`, `cor32`, `prop33`, `prop33_abs`, `absPi_lower`, `absPi_upper`, `absPi_le_kappa`, `absPi_lt_kappa`, `absPi_eq_zero_iff`, `absPi_strictAntiOn`, `absPi_equal`, `rcrit_iff`, `rcrit_eq_iff`, `rcrit_example`, `pc_lipschitz`, `thm35_bracket`, `prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`, `prop36_ar1_pcov`, `prop36_ar1_pvar`, `prop36_ar1`, `prop36_ar1_ne_zero`, `prop37_pcov`, `prop37_bound`, `prop38_ratio`, `prop38_loading`	population part checked

Modelling choices. Summary graphs are relations on a finite type with a self-loop hypothesis; anc and desc are reflexive-transitive closures ("a path exists iff a walk exists" is the one convention adopted without proof); G_T with $T = n + 1$ slices

lives on $V \times \mathrm{Fin}(n+1)$; source SCCs are represented by their member nodes (equality of SCC sets as equality of node unions is elementary and not formalised). In the finite-sample files, zero-mean variables are vectors of an arbitrary real inner-product space and lag-0 covariance is the inner product; independence enters only as vanishing inner products.

Findings. No statement was false. Formalisation added four missing nondegeneracy hypotheses to our earlier finite-sample statements: (i) the strict claims of Proposition 12 need $b_u b_v \neq 0$; (ii) $\ell_u = \rho_{uZ}^2 / r$ needs $r > 0$; (iii) nonvanishing of the multi-filter residual needs $b_u b_v \neq 0$; (iv) the partial correlation needs positive partial variances ($\mathrm{Var} Y_u > 0$ suffices). It also showed that one hypothesis of Proposition 4 and one of Lemma 19 are redundant, and that the in-forest statement needs acyclicity, not only out-degree at most one.

G. Real-Data Details

CausalRivers (Stein et al., 2025). Study 1 (P1–P3): the benchmark's connected-subgraph sampler, seeds 7000–7029, East Germany and Bavaria, $d \in \{5, 10\}$, 120 subgraphs, 791 induced edges; block means at 12 h and 24 h; nominal Fisher-z. Study 2 (P4–P6): seeds 8000–8029, 120 subgraphs, 1,376 non-relative pairs; proxy Z is the mean of standardised stations of the region outside the subgraph and outside 2 undirected river hops (429–615 stations); primary test on partial correlations with a residual Bartlett n_{eff} ($K = 100$ steps at 12 h, 50 at 24 h); median n_{eff} 139/113 (raw) and 231/211 (residual). Before conditioning, 0.98 (nominal) and 0.87 (effective) of non-relative pairs tested dependent. Median proxy contamination by river relatives was 2.6% (maximum 72%), a case of the contaminated-proxy term in Proposition 12. Post hoc and without verdict: conditioning on $k = 1, 3, 5, 10$ principal components of the leave-out stations gave dependent fractions $0.745, 0.605, 0.576, 0.491$ (12 h), with accuracy 0.580 and Spearman 0.16 at $k = 10$.

Causal Chambers (Gamella et al., 2025). Candidate selection (recorded before analysis) compared Causal Chambers, CausalRivers with measured precipitation (rejected: the driver reaches each gauge through its own catchment filter, so by Proposition 12 a scalar proxy cannot remove it), a semi-synthetic cross-sectional manufacturing benchmark (rejected: not real, not time series) and fMRI with physiological regressors (rejected: no ground truth). Wind tunnel: runs `actuators_random_walk_2–16` (run 1 excluded before analysis: truncated); 16 random variables, 26 edges, 63 non-relative pairs per run; block means over 10 rows, first differences, standardisation, $n = 999$; effective-n Fisher-z with Bartlett $K = 20$. Light tunnel: `actuators_white`, 20 variables, 39 edges, ten segments of 2,000 steps, levels. Predicted values come from the package's published ground-truth graph with constant parameters removed. Secondary analyses (no verdicts): nominal n gives slightly more false positives (0.048), as Proposition 10 predicts; levels instead of increments gave 0.054, contradicting our own secondary prediction that levels of random walks would produce many spurious dependences; without block means ($D = 1$) relative recovery drops to 0.18 and Spearman to 0.10. In the light tunnel, polariser angles act on two sensors through Malus' law, a nonlinearity with near-zero linear correlation; as pre-registered, these relative pairs were never detected, without affecting orientation.

H. Pre-Registration Timeline and Deviations

Times are commit timestamps relative to the first pre-registration commit (T_0); all events occurred on one day. Commit hashes are withheld for anonymity and will be provided with the de-anonymised version. Commit timestamps are set by the committing client; the hosting service's push records are the stronger evidence and can be supplied to the area chair on request.

time	event
$T_0 + 0{:}00$	plan 1 committed: targets T1–T3, predictions P1–P3, kill rule
$T_0 + 0{:}14$	theory of Sections 3 and 4, enumeration checks, operationalisation of P1–P3 (before data)
$T_0 + 0{:}17$	study 1 results (P2, P3 fail; kill rule triggered)
$T_0 + 1{:}05$	plan 2 committed: F1–F3 targets, Lean target, predictions P4–P6, kill rule
$T_0 + 1{:}18$–$1{:}31$	Lean formalisation of Section 3 and algebraic parts of Section 4
$T_0 + 1{:}38$	finite-sample theory, simulations, operationalisation of P4–P6 (before data)
$T_0 + 1{:}44$	study 2 results (P4, P6 fail; kill rule triggered)
$T_0 + 4{:}58$	plan 3 committed: candidate scoping, graph-only predictions, P7–P9 (before any dependence statistic)
$T_0 + 5{:}02$	study 3 results (P7, P8 pass; P9 fails; replication passes)
$T_0 + 5{:}14$–$5{:}24$	Lean formalisation of the finite-sample lemmas; errata
[PLACEHOLDER: E1/E2 times]	**[PLACEHOLDER: pre-registration and result commits of E1 and E2]**

Deviations (all recorded at the time). (1) Plan 1 said "theorem-predicted s_ρ" without naming the version; the stationary version was fixed in writing before data were loaded (the finite version would predict $> 1/2$ for every subgraph, including chains). (2) Self-loops are excluded from the predicted average because the estimator cannot orient them; this does not change which side of $1/2$ the prediction is on. (3) Only 6 of 120 study-1 subgraphs lacked a confluence; the plan set no minimum size for that group, and its trivial pass is not interpreted. (4) Plan 2 did not say which test defines "dependent" in P4; the effective-n test was fixed as primary before data, and the nominal test fails P4 as well. (5) In study 3, a planned triad-reliability check was not applicable (the measured driver reaches only four mutually related barometers) and is not interpreted. (6) The P9 failure is close to the threshold and its CI contains 0.3; as pre-registered, the point estimate decides.

I. Placeholders for Experiments E1 and E2

[PLACEHOLDER: E1 (discriminating test against naive predictors): design, pre-registration time relative to T_0, naive predictors compared, thresholds, full results table, deviations. To be filled from the experiment folder once the run and its README are committed. Do not fill with estimates.]

[PLACEHOLDER: E2 (lagged relatives): lag window, test, effective sample size, effect on the weak edges that limited P9, full results table, deviations. To be filled from the experiment folder once the run and its README are committed.]